

TASK5 REPORT (Data Analyst/ELEVATE LABS)

EDA Report on Titanic Dataset

Tools used: Python (Pandas, Matplotlib, Seaborn)

About dataset and the objectives of the analysis:

The **Titanic** dataset contains information about passengers who were on board the Titanic. This analysis explores the features of passengers and how they correlate with survival outcomes.

Dataset Overview:

The Titanic dataset consists of three CSV files:

1. **train.csv** – Contains the training data with features and the target variable (Survived).
2. **test.csv** – Contains the testing data with similar features but without the target.
3. **gender_submission.csv** – A sample submission file that shows what a correct submission format looks like.

Data Preprocessing

- Checked for null values.
- Used `.info()` and `.describe()` for initial understanding.
- Detected missing values in Age, Cabin, and Embarked columns.

Visualizations & Observations:

A. Survival Count

- Around 38% passengers survived.
- Data is slightly imbalanced with more non-survivors.

B. Survival by Gender

- Females had a significantly higher survival rate.
- “Women and children first” policy may have influenced this.

C. Survival by Passenger Class

- 1st class passengers had much better survival chances.
- 3rd class had the highest death rate.

D. Age Distribution by Survival

- Children (under 10) had a better survival rate.
- Many deaths occurred in 20–40 age range.

E. Fare vs Pclass

- 1st class fares were the highest.
- Some passengers paid very high fares (outliers).

F. Pairplot of Selected Features

- Fare and Pclass show strong patterns with survival.
- Good features for prediction modeling.

G. Age Distribution

- The peak age group is around 25–30 years.
- The data shows a decline in frequency as the age increases beyond 40.

H. Correlation Heatmap

- The heatmap shows the pairwise correlation between numerical variables in the Titanic dataset using Pearson correlation coefficients (values range from -1 to 1).
- Age has weak/no correlation with survival but may still provide useful nonlinear patterns when analysed by groups (like children vs adults).

Conclusion: The Exploratory Data Analysis (EDA) of the Titanic dataset provided valuable insights into the factors that influenced passenger survival during the tragedy.